MATH 128a: Homework #6

Instructor: F. Weber

Due by: **Wednesday, October 15, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 2.3

Solve problems 2, 6, 8, 14ac(18ac), 18(16) 30(19) of Section 2.3 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

You can use the Matlab codes on bcourses for problems 6, 8, 14, 18 if useful.

Problem 2: Problems from the textbook, chapter 2.4

Solve problems 1b, 6b, 10 of Section 2.4 of the textbook.

6b: Show that p_n converges at order $\alpha = 1$, but not linearly, since $\lambda = 1$ in Definition 2.7.

(If the problem has a different number in the 9th edition book, it is in parentheses)